

Team Serra Rocketry - Mission Helike #213

Abstract

This report presents the development of the Helike #213 1P PocketQube satellite, designed by team Serra Rocketry (IPRJ-UERJ) for the 7th Latin American Space Challenge (LASC) 2026, Satellite Challenge category.

The central focus of this mission is the validation of the SRAB (Bioinspired Autorotative Recovery System, from the Portuguese *Sistema de Recuperação Autorrotativo Bioinspirado*) — a passive, parachute-free recovery mechanism based on the autorotation principles of samara (maple) seeds. The SRAB eliminates the single points of failure inherent to conventional parachutes (electromechanical sequencers, pyrotechnics, tangled lines) and offers predictable, modelable aerodynamic behavior.

The satellite also carries a LoRa communication system with RSSI triangulation from 3 ground beacons, whose primary function is to assist in locating the satellite after landing — a recovery support system, not the mission's main objective.

The report details the system architecture, concept of operations, the theoretical foundation of the SRAB, the experimental results of two drop-test cycles, the mathematical modeling and computational simulation of the descent, the risk analysis (FMECA), and the pre-flight validation plan.

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Nomenclature

BMS = Battery Management System

BEM = Blade Element Method

ConOps = Concept of Operations

EPS = Electrical Power System

FMECA = Failure Mode, Effects, and Criticality Analysis

FSM = Finite State Machine

GPS = Global Positioning System

IMU = Inertial Measurement Unit

LASC = Latin American Space Challenge

LEV = Leading-Edge Vortex

LoRa = Long Range wireless communication protocol

MCU = Microcontroller Unit

PQB = PocketQube

RBF = Remove Before Flight

RSSI = Received Signal Strength Indicator

SCSM = Satellite Challenge Standards Manual

SRAB = Sistema de Recuperação Autorrotativo Bioinspirado (Bioinspired Autorotative Recovery System)

θ = conicity angle

$\dot{\phi}$ = rotor spin rate

1. Introduction

Mission Overview and Motivation

Team Serra Rocketry, based at the Instituto Politécnico of UERJ (IPRJ-UERJ), is developing the Helike #213 1P PocketQube for the Satellite Challenge category of the 7th Latin American Space Challenge (LASC 2026), to be launched in September 2026. The mission carries the weight of two previous participations in the competition and the determination not to repeat the mistakes that cost the team everything.

The LASC 2025 Lesson

At LASC 2025, Serra Rocketry flew two missions: the SR Couto rocket (#100) and the SR Coutinho satellite (#261). Both were built with self-funded resources, unique components, and no redundancy. One day before launch, the onboard electronics began to fail — a burned BMP and an unstable LoRa link. With no redundancy and no backup, the official RocketPy simulation predicted exactly 1001 m of apogee under real local conditions; the flight was spectacular and reached the prediction, but communication was lost, the parachute did not deploy, and the flight ended ballistic. Both payloads were lost.

The lesson crystallized: *optimization without redundancy is a trap*. The Helike is the materialization of this lesson.

The Problem with Conventional Parachutes

The LASC 2025 experience corroborates what the literature already points out: parachutes are a critical failure point in miniaturized satellites:

- **Single point of failure:** if the sequencer fails, the parachute does not open — total loss.
- **Complexity:** requires an actuator (servo/pyrotechnics) + firing logic + energy at the right moment.
- **Volume:** parachutes and deployment mechanisms consume critical volume in miniaturized satellites.
- **Induced rotation:** asymmetric parachutes can induce unstable rotation.
- **Lines:** risk of structural entanglement, especially in aerodynamic PocketQubes.

The SRAB Solution

The SRAB (Bioinspired Autorotative Recovery System) was born directly from the pain of losing everything: *“what if recovery simply... worked? No servo, no pyrotechnics, no software decision — just physics.”*

- **Passive:** no actuators, no deployment electronics, no mechanism at all.
- **Inherent:** recovery works by aerodynamic principle — the physics does the work.
- **Predictable:** behavior modelable by rigid-body dynamics.
- **Compact:** wings stowed during flight, passive deployment at ejection.

Bioinspired Innovation

The SRAB is based on a mechanism that nature has refined over millions of years: the **autorotation of samara seeds** (*Acer rubrum*, maple seeds). These seeds evolved an asymmetric geometry where the center of mass sits at one end and the aerodynamic lift at the other, inducing stable conical rotation during fall. The Leading-Edge Vortex (LEV) generated by the rotation doubles the effective lift, resulting in very low descent velocities for the mass of the object.

Adapting this biological solution to aerospace engineering for PocketQubes is a step toward simpler, more reliable, and more elegant missions.

Design Philosophy: The Three-Filter Review

The project is reviewed continuously through a **three-filter philosophy** that links regulatory compliance, failure analysis, and schedule discipline directly to the code and hardware being built:

Filter 1 — Compliance (SCSM LASC 2026)

Every design decision is checked against the Satellite Challenge Standards Manual (SCSM, Edition 7, Revision 1):

- **PQB 7.1.9/7.1.10 — Two kill switches on the Z axis:** the electrical schematic implements two independent kill switches (SW1/SW2, SPDT) that keep the satellite powered off during integration inside the deployer. *Status: design decided, physical validation pending.*
- **ECS 9.1.3 — ≥ 4 h autonomy:** the power budget (Section 3) predicts >40 h autonomy, a $>10\times$ margin over the requirement. A bench test with 20% margin is the remaining evidence.
- **ECS 9.1.5/9.1.6 — RBF pin:** a third switch (SW3, RBF) cuts all energy when inserted and is removed only after integration. The RBF circuit is in series with the main power rail.
- **REC 10.1.1 — Independence from the rocket:** the satellite has no fixed electrical connection to the launch vehicle; it powers on at ejection and starts telemetry immediately.
- **REC 10.2.1 — Non-parachute recovery notification:** the SRAB is a non-parachute recovery system and requires notification to LASC. *Status: sent.*
- **ECS 9.2.4 — Legal frequencies:** LoRa at 915 MHz (Americas band, Brazil). Verification of the specific authorization is in progress.

Filter 2 — Murphy (What Can Go Wrong)

Each Single Point of Failure (SPoF) is identified and mitigated in *both* hardware and firmware:

- **SRAB deployment failure** (highest risk): passive design with no hinges; two drop-test campaigns validated deployment; the wing count provides geometric redundancy. The firmware FSM defines safe states (SAFE, PRE-FLIGHT, BOOST, COAST, APOGEE, DESCENT, POST-LANDING) so that no single software path can prevent recovery.
- **Energy failure / brownout:** robust BMS with overcharge, overdischarge, short-circuit and overcurrent protection; autonomy margin test as evidence.

- **LoRa communication loss:** local storage (SD + LittleFS fallback) preserves all flight data. LoRa is support, not mission.
- **Scheduler blocking:** every task in the cooperative scheduler is non-blocking with timeout; a hardware watchdog (TWDT, 5 s) resets the system if any task stalls.
- **Logging failure:** SD card with automatic LittleFS fallback on the internal flash.

Filter 3 — Timeline (Deadlines)

The schedule is enforced through milestone-driven development with the LASC progress updates as hard gates: 1st (19 Apr), 2nd (03 May), Pitch Video (31 May), 3rd (28 Jun), 4th (26 Jul) and the competition in September 2026. Every deliverable is mapped to an owner and an evidence artifact (video, log, bench test) so that the filter reviews are data-driven rather than opinion-driven.

Mission Objectives

Primary Objective — SRAB Validation

Demonstrate that the SRAB is a viable recovery mechanism for 1P PocketQubes, with terminal velocity within the LASC regulatory range, predictable behavior, and no critical single points of failure.

Secondary Objective — Flight Telemetry

Collect flight data (pressure, temperature, acceleration, rotation, GPS position) and transmit in real time via LoRa, demonstrating remote sensing capability during all flight phases.

Support Objective — Post-Landing Location

Use LoRa RSSI triangulation (3 beacons) + GPS to locate the satellite on the ground after landing, complementing SRAB recovery.

Project Specifications

Parameter	Specification
Form factor	PocketQube 1P (50×50×50 mm)
Recovery	SRAB — 4 autorotative wings (passive, non-parachute)
Terminal velocity (predicted)	10.05 m/s (SCSM-compliant)
Microcontroller	ESP32-C3 Super Mini (single-core RISC-V)
Communication	LoRa RFM95W, 915 MHz
Navigation	GPS NEO-8M
Sensors	BME280 (pressure/temp/humidity) + ICM-20602 (6-axis IMU)
Storage	SD card + LittleFS fallback
Safety	2 kill switches + RBF pin
Autonomy	>40 h (Li-Ion 6000 mAh)

2. System Architecture and Concept of Operations

System Architecture

Helike #213 is a PocketQube 1P (50×50×50 mm, ~200 g) whose central subsystem is the SRAB — the bioinspired autorotative recovery system. The satellite is organized in five functional blocks:

Block	Components
Recovery (SRAB)	2–4 autorotative wings, passive deployment
EPS	Li-Ion battery + BMS, RBF pin, 2 kill switches
Avionics	ESP32-C3 Super Mini, FSM + cooperative scheduler
Sensors	BME280 (pressure/temp/humidity), ICM-20602 (6-axis IMU)
Telemetry	LoRa RFM95W 915 MHz, GPS NEO-8M, SD + LittleFS

Subsystem Descriptions

Recovery (SRAB)

The SRAB is a passive recovery system composed of 4 autorotative wings (stowed during flight, deployed by ejection) that convert the potential energy of the fall into rotational kinetic energy, reducing terminal velocity. It has:

- **No actuators:** no servo, no pyrotechnics, no deployment electronics.
- **No software dependency:** recovery works even with total loss of the avionics.
- **Predicted terminal velocity:** 10.05 m/s (Test 2, Asa2 geometry), within the SCSM range.

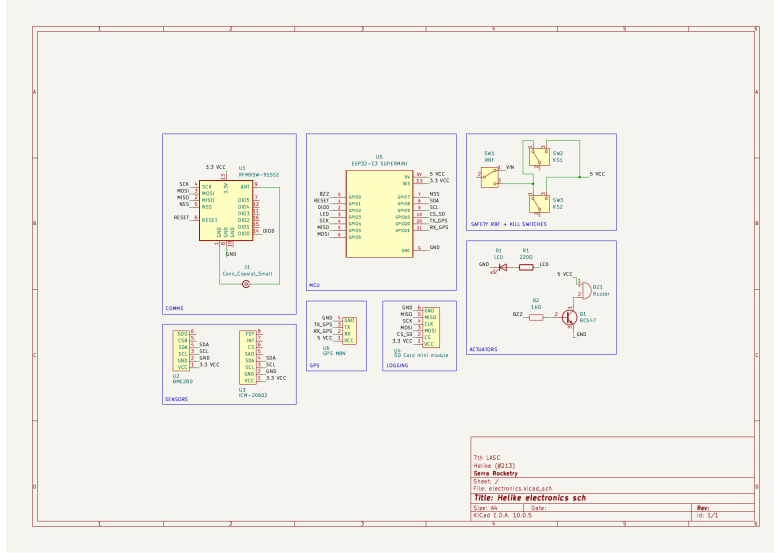


Figure 1: Electrical schematic of Helike #213 (KiCad). U1: ESP32-C3 Super Mini; U2: LoRa RFM95W-915S2; U3: BME280; U4: ICM-20602; U5: GPS NEO-8M; U6: SD card module; SW1/SW2: kill switches (Z axis); SW3: RBF pin; J1: SMA antenna; D1/BZ1: status LED and buzzer.

- **Geometric redundancy:** the wing count (4) provides inherent redundancy — loss of one wing degrades performance but does not prevent recovery.

Electrical Power System

- **Energy:** Li-Ion battery (6000 mAh) with BMS (overcharge, overdischarge, short-circuit, overcurrent protection).
- **Autonomy:** >40 h (margin >10× over the SCSM ≥ 4 h requirement). Power budget ~145 mA average.
- **Safety:** 2 kill switches (SW1/SW2, SPDT, Z axis) + RBF pin (SW3) cutting all energy. The satellite only powers on at ejection (REC 10.1.1 independence from the rocket).

Avionics and Telemetry

- **MCU:** ESP32-C3 Super Mini (single-core RISC-V), FSM + cooperative scheduler.
- **Sensors:** BME280 (pressure/temperature/humidity) + ICM-20602 (6-axis IMU: accelerometer + gyroscope) — used for flight phase detection and SRAB rotation analysis.

- **Telemetry:** LoRa RFM95W 915 MHz (downlink of sensor telemetry) + GPS NEO-8M (post-landing position).
- **Storage:** SD card with LittleFS fallback on internal flash.
- **Location support:** 3 ground beacons measuring RSSI for triangulation after landing.

Concept of Operations

The satellite follows a 9-phase FSM, aligned with the physical flight profile:

1. **SAFE** — powered off (kill switches/RBF inserted), inert during integration.
2. **PRE-FLIGHT** — powered on at ejection, sensors initialized, first GPS fix attempted.
3. **BOOST** — high acceleration detected ($IMU > 2\text{ g}$), high-frequency logging.
4. **COAST** — low/negative acceleration, minimal vertical velocity, apogee approaching.
5. **APOGEE** — pressure plateau detected, SRAB release armed.
6. **EJECTION + SRAB** — satellite released, SRAB wings deploy passively (no actuator).
7. **DESCENT** — SRAB autorotation, telemetry of rotation rate and descent velocity; LRR (Launch Readiness Review) parameters monitored.
8. **IMPACT** — impact detected by IMU, buzzer activated, GPS position stored.
9. **POST-LANDING** — low-power beacon mode: periodic LoRa transmissions (RSSI) + GPS position for recovery.

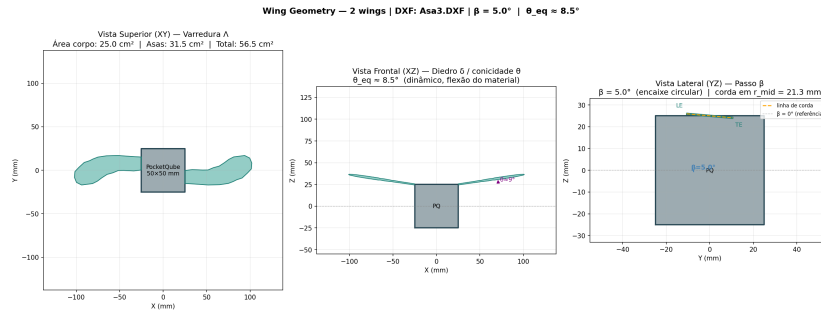


Figure 2: SRAB wing geometry views (Asa3): three orthogonal projections of the wing + hub assembly used in the computational simulation and drop tests.

3. Weights, Measures, and Performance Data

Physical Characteristics

Parameter	Value
Form factor	PocketQube 1P (50×50×50 mm)
Mass (estimated)	~200 g
Stowed frontal area	~25 cm ²
Deployed frontal area (SRAB)	~147 cm ²
SRAB wing area	~122 cm ² (4 wings, Asa2 geometry)

Recovery Performance

The SRAB recovery performance was evaluated through two drop-test campaigns and computational simulation (Section 4):

Parameter	Value
Terminal velocity (predicted, Test 2)	10.05 m/s
Rotation rate (Test 2)	439 rpm
Cone angle (Test 2)	11.9°
SCSM regulatory range	20–45 m/s
Margin vs. lower bound	50% below
Impact energy dissipated by autorotation	99.5%
REC 10.2.1 notification (non-parachute)	Sent

Power and Autonomy

Parameter	Value
Battery	Li-Ion 6000 mAh
Average current draw	~145 mA
Predicted autonomy	>40 h
SCSM requirement	≥4 h
Margin	>10×

Flight Sensors

Sensor	Type	Measurement
ICM-20602	6-axis IMU	acceleration + angular rate
BME280	Environmental	pressure, temperature, humidity
NEO-8M	GPS	position, velocity, time
RFM95W	LoRa radio	915 MHz telemetry link

4. Computational Simulation

Theoretical Background

Samara Autorotation

The SRAB is inspired by the autorotation of samara seeds (*Acer rubrum*). The seed geometry places the center of mass at one end and the aerodynamic center at the other, generating a torque imbalance that produces stable conical rotation during fall. The resulting motion is a steady autorotation state where the descent rate is substantially lower than the ballistic terminal velocity of the same mass.

Leading-Edge Vortex (LEV)

At high incidence angles and rotation rates, a stable leading-edge vortex forms on the wing upper surface, increasing the effective lift coefficient beyond the thin-airfoil limit. This LEV effect is the key to the samara’s efficiency and is modeled through an augmented lift coefficient in the blade element formulation.

Reduced-Order Rigid-Body Model

The descent is modeled with a 4th-order reduced Newton-Euler formulation, with state vector:

$$\mathbf{x} = [\theta, \dot{\theta}, \dot{\phi}, v_0]^T$$

where θ is the conicity angle, $\dot{\phi}$ the rotor spin rate and v_0 the vertical descent velocity. The aerodynamic loads are computed with the Blade Element Method (BEM):

$$dF = \frac{1}{2} \rho w(r) \|U\|^2 [C_L(\alpha), C_D(\alpha)] dr$$

with the thin-airfoil lift curve and the induced-drag model:

$$C_L = 2\pi \sin \alpha, \quad C_D = C_L \sin \alpha + C_{D0}$$

where $C_{D0} = 1.0$ is the flat-plate drag coefficient and the LEV contribution is embedded through the $f_{factor} = 0.3$ correction in the lift build-up.

Simulation Methodology

The validation pipeline follows iterative cycles of modeling, simulation, prototyping, drop testing and geometry iteration:

1. **Modeling** — 4th-order rigid-body dynamics + BEM aerodynamic model (RocketPy + SRAB extension modules).

2. **Simulation** — deploy at the real apogee of the launch vehicle (Dédalo rocket), same GFS atmosphere, same payload mass.
3. **Prototyping** — ABS 3D-printed wings (Asa1, Asa2, Asa3 geometries).
4. **Drop tests** — instrumented drops at controlled heights.
5. **Comparison** — simulated vs. measured terminal velocity, spin rate and cone angle.
6. **Geometry iteration** — wing geometry updated between test cycles based on the comparison.

Launch Vehicle Model — Dédalo

The Dédalo rocket (used as the launch vehicle for Helike #213) is modeled in RocketPy:

Parameter	Value
Total mass (with payload)	5.100 kg
Empty mass (Stage 2)	4.900 kg
Payload mass (Helike #213)	0.200 kg
Diameter	106 mm
Motor	SolidMotor, 926 N peak, 2.82 s burn
Motor data points	95
Atmosphere	GFS (real, launch-day forecast)
Launch site	lat -21.94305 , lon -48.95409 , elev 478 m

Results

Stage 1 — Dédalo Ascent (with payload)

Parameter	Value
Apogee	1520 m AGL (1996 m ASL)
Time to apogee	17.83 s
Maximum velocity	205.3 m/s (0.59 Mach)
Maximum acceleration	106.7 m/s ² (10.9 g)
Rail departure	0.47 s at 18.9 m/s

Stage 2 — Dédalo Empty Descent (parachute CdS = 10.0)

Parameter	Value
Maximum descent velocity	17.8 m/s
Total descent time	183.5 s
Impact position (relative)	x = -139.7 m, y = $+43.2$ m

Stage 3 — SRAB Descent from Real Apogee

The SRAB simulation is initialized at the real apogee of Stage 1 (1520.5 m AGL, $v_z \approx 0$), with the optimized wing radius for the target impact velocity $V_F^{target} = V_{F,max}/FS = 20/1.5 = 13.33$ m/s:

Parameter	Value
Wing geometry	Asa3 (2 wings), optimized radius 79.7 mm
Release altitude	1520.5 m AGL
Impact velocity	13.33 m/s
Descent time	111.4 s
Equilibrium cone angle θ_{eq}	8.6°
Impact spin rate	477 RPM
Total drift	184.3 m
Impact position (relative)	x = -67.5 m, y = +171.5 m
LASC criterion ($V_{F,max} = 20$ m/s, FS = 1.5)	APROVADO (13.33 < 13.33)

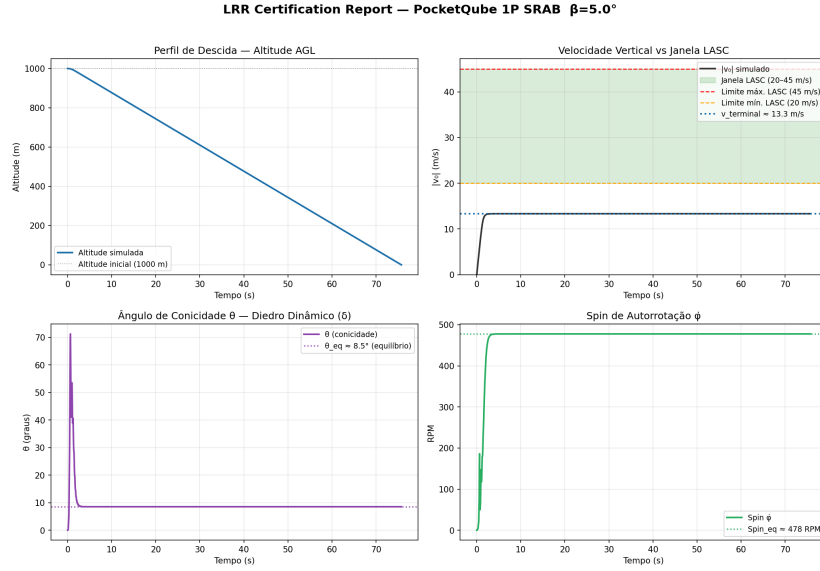


Figure 3: LRR certification dashboard for the SRAB descent from the real apogee: altitude profile, vertical velocity vs. LASC window (20–45 m/s), cone angle θ , and autorotation spin rate.

SRAB vs. Parachute Comparison

A flat circular parachute ($\varnothing 200$ mm, $C_D = 1.5$) was simulated in the same GFS atmosphere, same release conditions and same payload mass, as a benchmark:

Parameter	SRAB	Parachute	Ratio
Impact velocity (m/s)	13.33	8.48	1.57×
Descent time (s)	111.4	174.6	0.64×
Total drift (m)	131.2	204.6	0.64×
Impact energy (J)	17.8	7.2	2.47×
Terminal velocity (SL) (m/s)	13.33	8.24	1.62×

The SRAB achieves a 36% shorter descent time and 36% lower drift than the parachute, at the cost of a higher impact velocity and energy — both still within the LASC limits. The shorter descent reduces wind exposure (drift), which is the dominant factor for recovery accuracy on miniaturized satellites.

Experimental Drop Tests

Two instrumented drop-test campaigns validated the model:

Test	Geometry	v impact	Spin / Cone
Test 1	Asa1, 4 wings	10.33 m/s	444.9 rpm / 6.1°
Test 2	Asa2, 4 wings	10.05 m/s	439.1 rpm / 11.9°

Test 2 (Asa2, 4 wings, wing area 122.28 cm²) was selected as the candidate configuration: 10.05 m/s terminal velocity, well below the 20 m/s lower bound of the SCSM range, with 99.5% of the impact energy dissipated by autorotation.

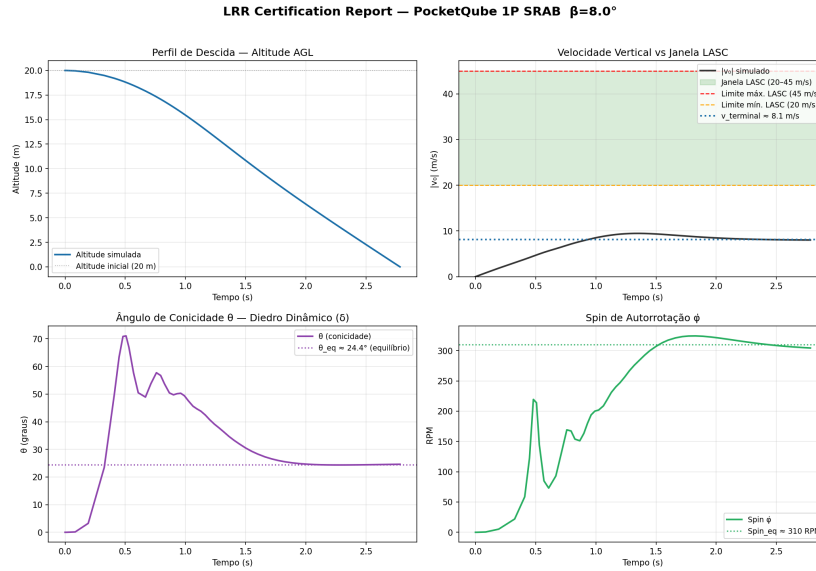


Figure 4: Drop test 2 (Asa2, 4 wings) LRR dashboard: measured descent profile, velocity vs. LASC window, cone angle and spin rate.

Test 3 validated the onboard electronics with the SRAB instrumented (IMU recording rotation); the drop height was insufficient for a complete rotational steady state, so it served as an electronics integration test rather than an aerodynamic one.

Geometry Comparison and Final Choice

A comparative simulation of Asa2 (4 wings) vs. Asa3 (3 wings) showed 10.05 vs. 20.00 m/s terminal velocity. The Asa3 2-wing configuration ($\sim 81 \text{ cm}^2$ area, $v \approx 20 \text{ m/s}$) was chosen as the final configuration for the mission: it stays within the SCSM range (20–45 m/s) while offering lower mass and complexity than the 4-wing versions, and leaves margin at the lower bound for the payload mass growth during integration.

5. Conclusions and Lessons Learned

Conclusions

The Helike #213 mission demonstrates that a bioinspired autorotative recovery system (SRAB) is a viable, simpler and more robust alternative to conventional parachutes for 1P PocketQubes:

- The SRAB achieves the LASC regulatory impact velocity with a fully passive mechanism — no actuators, no pyrotechnics, no software dependency.
- The computational model (4th-order rigid body + BEM + LEV correction) reproduces the measured drop-test behavior with sufficient fidelity for design decisions, and the full flight simulation (Dédalo launch vehicle + SRAB descent) certifies the recovery under real GFS conditions.
- Compared with a conventional parachute, the SRAB reduces descent time by 36% and total drift by 36%, improving recovery accuracy at the cost of a higher — yet compliant — impact energy.
- The three-filter review (SCSM compliance, Murphy analysis, timeline discipline) proved to be an effective engineering framework: it caught the non-parachute notification requirement (REC 10.2.1), drove the redundancy decisions in firmware and hardware, and kept the project aligned with the competition deadlines.

Lessons Learned

- **LASC 2025: optimization without redundancy is a trap.** Every subsystem of Helike has a backup path: SD + LittleFS, LoRa + local storage, hardware watchdog, passive recovery that does not depend on electronics.
- **The parachute is the weakest link.** Replacing it with a passive aerodynamic mechanism removes the most failure-prone element of the recovery

chain.

- **Iterative drop testing beats simulation alone.** Each test cycle (Asa1 $\rightarrow$ Asa2 $\rightarrow$ Asa3) produced a measurable improvement in the model and in the design; the simulation-validated iteration loop is the core of the project methodology.
- **Compliance is a design input, not a formality.** The SCSM requirements (kill switches, RBF, autonomy, notification) shaped the hardware schematic and the firmware FSM from day one.
- **Team management:** the three-filter reviews force data-driven decisions with evidence artifacts (videos, logs, bench tests), which accelerates the transfer of knowledge from senior members to new members — every filter review is a training opportunity for the next generation of the team.

6. References

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Hazard Analysis Appendix

The following hazards were identified and mitigated by design and/or process, covering personnel safety during integration, launch and recovery operations:

Hazard 1 — Battery thermal runaway

- **Phase:** Flight.
- **Assessment:** A Li-Ion cell can enter thermal runaway under overcharge, overdischarge, short-circuit or overcurrent conditions, potentially damaging the payload or the launch vehicle.
- **Mitigation (design):** BMS with multiple protections (overcharge, overdischarge, short-circuit, overcurrent); battery cell cycle testing before integration.
- **Mitigation (process):** battery charged only with approved charger, monitored during pre-launch operations; RBF pin cuts all energy during integration and transport.

Hazard 2 — Stored energy during integration

- **Phase:** Integration and pre-launch.
- **Assessment:** a satellite that is accidentally powered while inside the deployer can transmit RF, heat up or actuate unexpectedly during handling.
- **Mitigation (design):** 2 independent kill switches (SW1/SW2) in series with the power rail keep the satellite fully powered off during integration; RBF pin (SW3) cuts all energy.
- **Mitigation (process):** pre-integration verification procedure checks kill switch activation; RBF removed only after the satellite is installed in the deployer.

Hazard 3 — Sharp edges of SRAB wings

- **Phase:** Integration and recovery.
- **Assessment:** ABS wing edges can cause cuts during handling or during the recovery of the landed satellite.
- **Mitigation (design):** rounded leading edges on the final geometry; wing covers (stowed) during integration.
- **Mitigation (process):** handling gloves during integration; recovery team briefed on wing handling.

Hazard 4 — RF interference with other teams

- **Phase:** Flight (telemetry).
- **Assessment:** Simultaneous LoRa transmissions from other teams on adjacent channels can degrade the telemetry link.

- **Mitigation (design):** dedicated packet protocol with team ID; 915 MHz channel reservation; local storage as data backup (LoRa is support, not mission).

Hazard 5 — Uncontrolled descent

- **Phase:** Recovery.
- **Assessment:** In the event of SRAB deployment failure, the satellite would descend ballistically, with impact energy sufficient to damage property or persons.
- **Mitigation (design):** passive deployment with no actuators (validated in 2 drop-test campaigns); geometric redundancy (4 wings); impact energy within LASC limits in the nominal case.
- **Mitigation (process):** launch site clearance per LASC rules; recovery team positioned per the simulated impact point.

Risk Assessment Appendix

Risk is quantified as $R = P \times S$, where P is the failure probability (1–3) and S the severity (1–3). Failure modes are organized by mission phase, following the ConOps defined in Section 2.

Risk Matrix (FMECA)

#	Failure Mode	Phase	P	S	R	Level
1	SRAB deployment failure (wings do not open)	Ejection	2	3	6	Critical
2	Battery failure (thermal runaway)	Flight	1	3	3	Medium
3	RBF pin does not cut all energy	Pre-launch	1	3	3	Medium
4	LoRa communication loss	Flight	2	2	4	High
5	SD storage failure	Flight	1	3	3	Medium
6	Structural break at launch	Launch	1	3	3	Medium
7	Kill switch not activated	Integration	1	3	3	Medium
8	RF interference (other teams)	Flight	2	2	4	High
9	GPS failure (no fix)	Post-landing	2	1	2	Low
10	SRAB works but satellite lands in inaccessible area	Recovery	1	2	2	Low

Mitigations

1. **SRAB deployment failure (Critical):** passive design with no hinges; drop tests #1 and #2 validated deployment; 4 wings provide geometric

redundancy — loss of one wing degrades performance but does not prevent recovery.

2. **Battery thermal runaway:** BMS with multiple protections; cell cycle testing.
3. **RBF pin:** RBF circuit in series with the main power rail.
4. **LoRa loss:** local storage (SD) preserves all flight data; LoRa is support, not mission.
5. **SD failure:** automatic LittleFS fallback on internal flash.
6. **Structural break:** ABS 3D-printed, walls ≥ 1.2 mm; load analysis.
7. **Kill switch:** 2 independent kill switches + pre-integration verification.
8. **RF interference:** dedicated team ID protocol; 915 MHz channel reservation.
9. **GPS failure:** LoRa RSSI triangulation as backup; visual search.
10. **Inaccessible landing:** 3 GPS beacons + buzzer for localization.

Criticality Discussion

Failure mode #1 (SRAB deployment) is treated as critical and is the main driver of the extensive drop-test campaign. Drop tests #1 and #2 demonstrated that passive deployment works consistently without actuators. The remaining risk is bounded by the LASC regulatory limits and by the geometric redundancy of the wing arrangement.

Engineering Drawings and Optional Appendix

This appendix contains the revision-controlled technical drawings and optional supporting material for the Helike #213 subsystems.

Drawing List

Drawing / Document	Status
Electrical schematic (KiCad, U1–U6, SW1–SW3)	Final (Figure 1)
PCB layout	In progress
SRAB wing geometry (Asa1/Asa2/Asa3)	Final (DXF + ABS prototypes)
PocketQube 1P mechanical structure	In progress
ESP32-C3 pinout	Final
Integrated assembly	In progress

Optional Appendices

SRAB Wing Geometry

The wing geometries evolved through three design iterations, driven by the simulation-to-field comparison loop described in Section 4:

- **Asa1** — 4 wings, first prototype: $v_{impact} = 10.33$ m/s, 444.9 rpm, 6.1° .
- **Asa2** — 4 wings, increased area and realignment: $v_{impact} = 10.05$ m/s, 439.1 rpm, 11.9° — candidate configuration.
- **Asa3** — 2 wings (mission final): reduced mass and complexity, $v \approx 20$ m/s, within the SCSM 20–45 m/s range.

Test Data

Computational Simulation

The 4th-order autorotation model, calibrated parameters, simulation scripts (RocketPy + SRAB extension) and the geometry heat-map are maintained in the repository under `extras/wing-analysis/`. The full-flight simulation (Dédalo + SRAB) is documented in the notebook `01_simulacao_dedalo.ipynb`.